

Initial Post

The power of descriptive statistics lies in the skilful and concise summarisation of a dataset's characteristics, enabling a holistic understanding of the data without making predictions or generalising the findings (Dong, 2023). A key aspect of data analysis is data visualisation, which, following the Occam's razor principle, favours a simplicity-first approach (Bader and Leuzinger, 2025). This leads to a clear and powerful visualisation technique that helps readers quickly understand the key message and support the decision-making process.

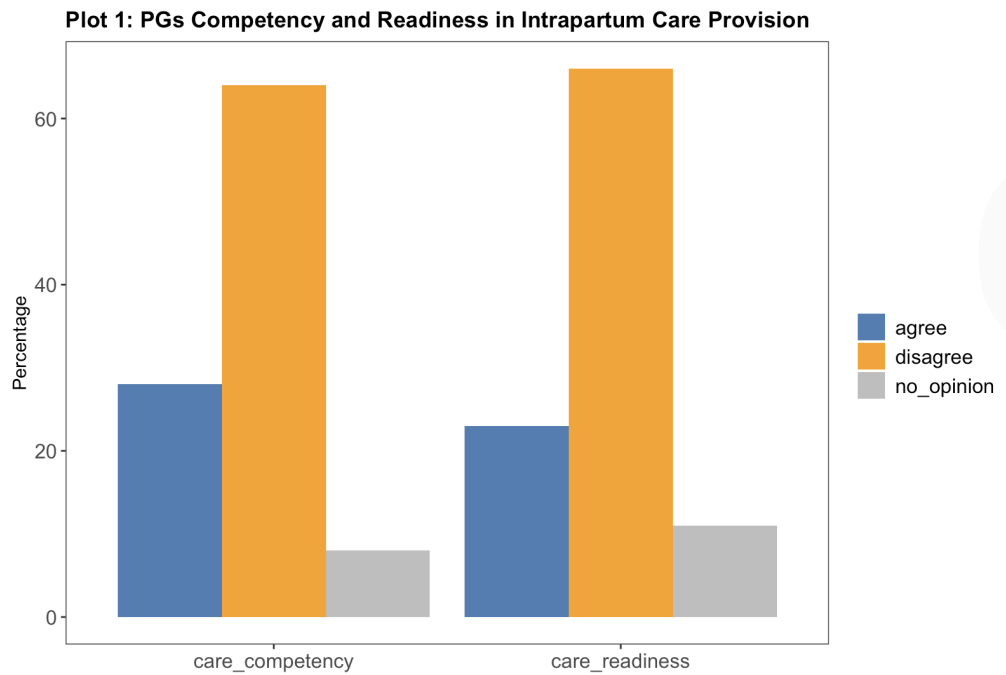
Activity and Approach

The purpose of this activity was to succinctly present the survey results shown in Table 2 of Brown's article (Brown, 1994) and to provide visual support for the author's findings concerning the provision of intrapartum care.

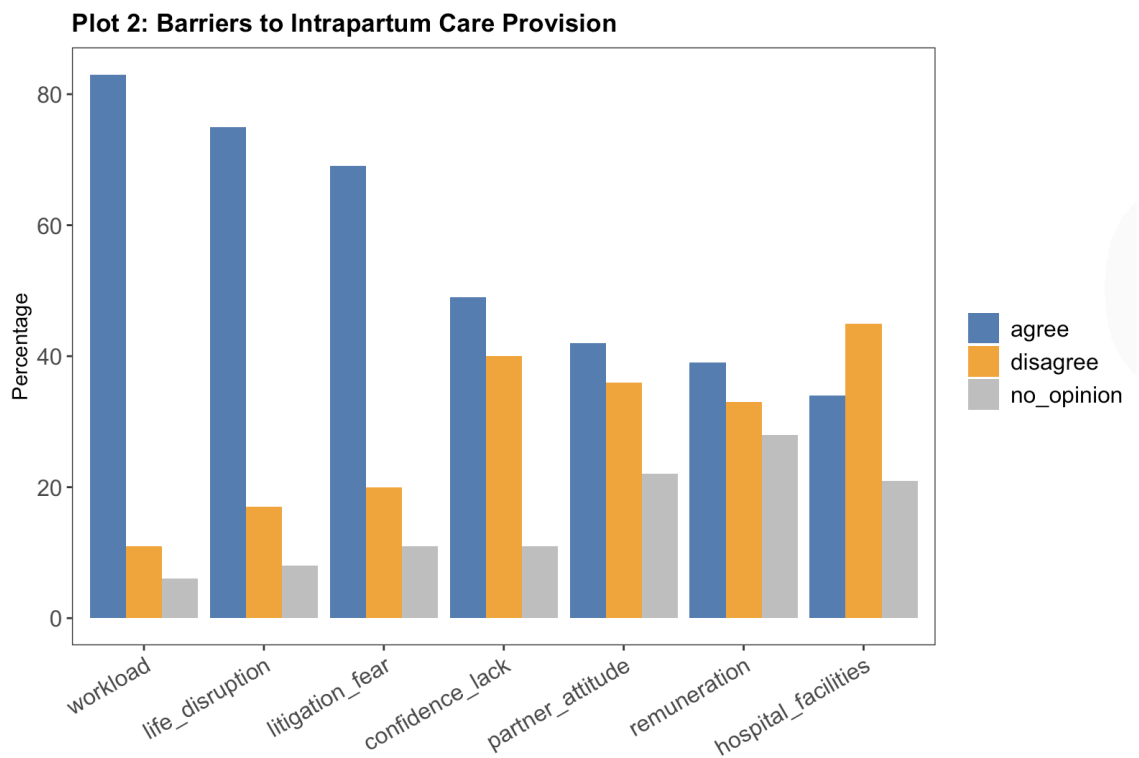
The survey, which consists of 24 statements, collects responses from 550 general practitioners (GPs) and aggregates them on a 3-point Likert scale (Agree, Disagree, No opinion). The survey results were organised into three tables (see Appendix) representing different insights, each visualised as a corresponding bar plot.

Findings

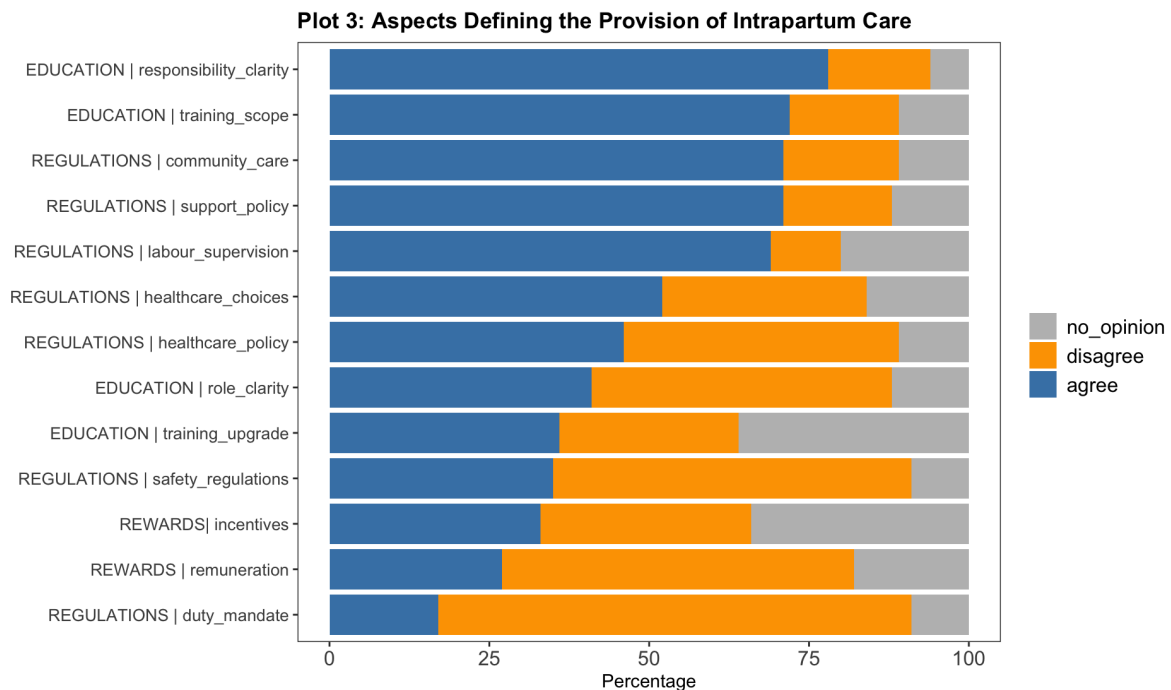
Plot 1 shows that readiness for and competency in intrapartum care have similar response distributions. This indicates that practitioners who feel less competent (64%) are also less likely to provide adequate care (66%).



Further, Plot 2 highlights barriers to offering intrapartum care, with the main issues being the high workload of medical practitioners (83%), the impact on their private lives (75%), and fear of litigation (69%). In contrast, monetary incentives (39%) and available hospital care facilities (33%) are seen more as secondary factors when providing care.



Additionally, Plot 3 shows that educational and training opportunities, regulatory aspects, and the rewards system are emerging as critical indicators of practitioners' competence and readiness to support women interested in intrapartum care.



In particular, transparency regarding the responsibilities of care providers (78%), clarity about the scope of training (72%), women-centred service policies (71%), and support offered to them through community proximity (71%) are seen as central to providing appropriate intrapartum care. Practitioners also emphasise the importance of aligning supervision rules for their work (69%) and offering flexible healthcare choices for women (52%). Finally, although to a lesser extent, the rewards system should be reconsidered to improve compensation (33%) and incentives (27%) for GPs who provide these essential services.

Learnings and Reflection

The power of visualisation lies in delivering a message that is both simple and clear, leaving little room for misinterpretation. To capture the essence of lengthy survey questions and accurately reflect PGs' responses, I divided the data into smaller, self-contained subsets, provided concise yet representative labels, and grouped them thoughtfully into categories to deliver a focused analysis. Furthermore, the exercise required thoughtful, time-consuming consideration of what deserves visual

emphasis and how best to present it without diluting the message. Additionally, maintaining clarity when describing the plots, while avoiding generalisations or implying correlations between variables, was essential for communicating GPs' perceptions and their provision of intrapartum care.

References

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Appendix

Plot1_Competency_Care			
reason	agree	disagree	no_opinion
care_readiness	23	66	11
care_competency	28	64	8

Plot2_Reasons no providi...			
reason	agree	disagree	no_opinion
workload	83	11	6
life_disruption	75	17	8
litigation_fear	69	20	11
confidence_lack	49	40	11
partner_attitude	42	36	22
remuneration	39	33	28
hospital_facilities	34	45	21

Plot 3_Care requirements					
category	requirements	category requirements	agree	disagree	no_opinion
Education	responsibility_clarity	EDUCATION responsibility_clarity	78	16	6
Education	training_scope	EDUCATION training_scope	72	17	11
Education	role_clarity	EDUCATION role_clarity	41	47	12
Education	training_upgrade	EDUCATION training_upgrade	36	28	36
Regulations	support_policy	REGULATIONS support_policy	71	17	12
Regulations	community_care	REGULATIONS community_care	71	18	11
Regulations	labour_supervision	REGULATIONS labour_supervision	69	11	20
Regulations	healthcare_choices	REGULATIONS healthcare_choices	52	32	16
Regulations	healthcare_policy	REGULATIONS healthcare_policy	46	43	11
Regulations	safety_regulations	REGULATIONS safety_regulations	35	56	9
Regulations	duty_mandate	REGULATIONS duty_mandate	17	74	9
Rewards	incentives	REWARDS incentives	33	33	34
Rewards	remuneration	REWARDS remuneration	27	55	18

Summary Post

My initial post addressed editors' feedback on Brown's Table II presentation (Brown, 1994), which was found overwhelming and difficult for readers to comprehend—a view I share. To address this, I grouped the survey findings and clarified the narrative using a three-layered approach: first, leading the reader through the overall sentiment, then exploring the contextual reasons behind it, and finally closing with the three-dimensional aspects influencing the care provided.

As peer feedback confirmed, effective data analysis and visualisation is not about preserving all information at all costs, but about retaining enough insights to support the key message. Peers rightly noted that redistributing information does not necessarily mean reducing it. Therefore, my initial concerns about losing information were unwarranted. Building on this understanding, I now believe it is crucial to step back, determine what is truly critical, and decide what deserves to be extracted and highlighted for the audience. At the same time, I remain mindful of the potential risks of data misinterpretation or analytical bias, as Balan (2024) cautions. In particular, for my last graph, carefully filtering out only the information that warrants visualisation, or a table view, would reduce ambiguity, strengthen insights, and lessen cognitive load for readers.

Moreover, peer feedback offered valuable insights into choosing a visualisation approach, underscoring that graphic choices can draw attention to key insights and aid reader interpretation. Peers' visualisation selection for Likert-scale data ranged from the popular diverging stacked bars (Heiberger and Robbins, 2014) to traditional stacked bars and classic bar charts. However, I argue that choosing the right visualisation is not always straightforward. As Franconeri *et al.* (2021) emphasise, a visualisation misaligned with the audience can create interpretive challenges, dilute information, and reduce confidence in the analysis. Therefore, understanding the audience's needs, goals, and graphic literacy is essential for effective data presentation.

Overall, this exercise and the peer discussion deepened my appreciation for the power of effective visualisation. Careful data selection is essential, and not every finding merits visualisation. The goal is to create clear communication and insights

that foster understanding, help the intended audience identify what is critical, and support informed decisions, all while preserving enough context to build trust in the analysis.

References

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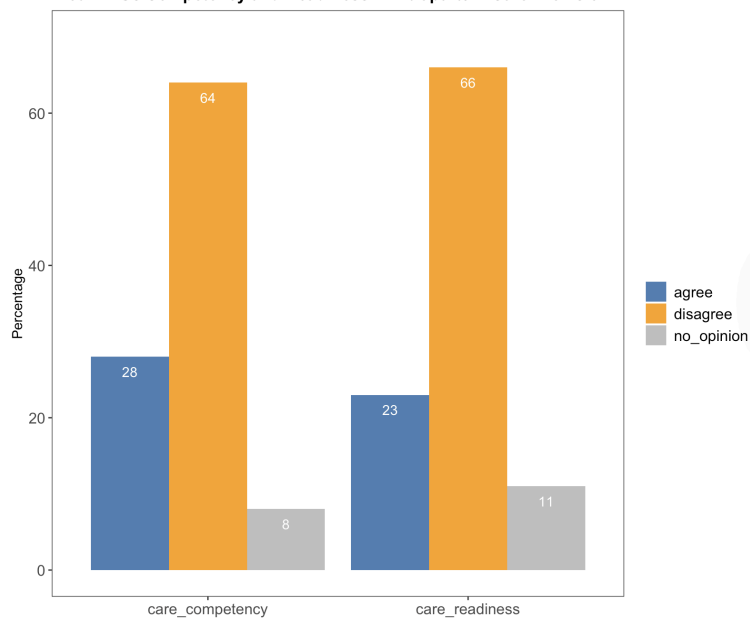
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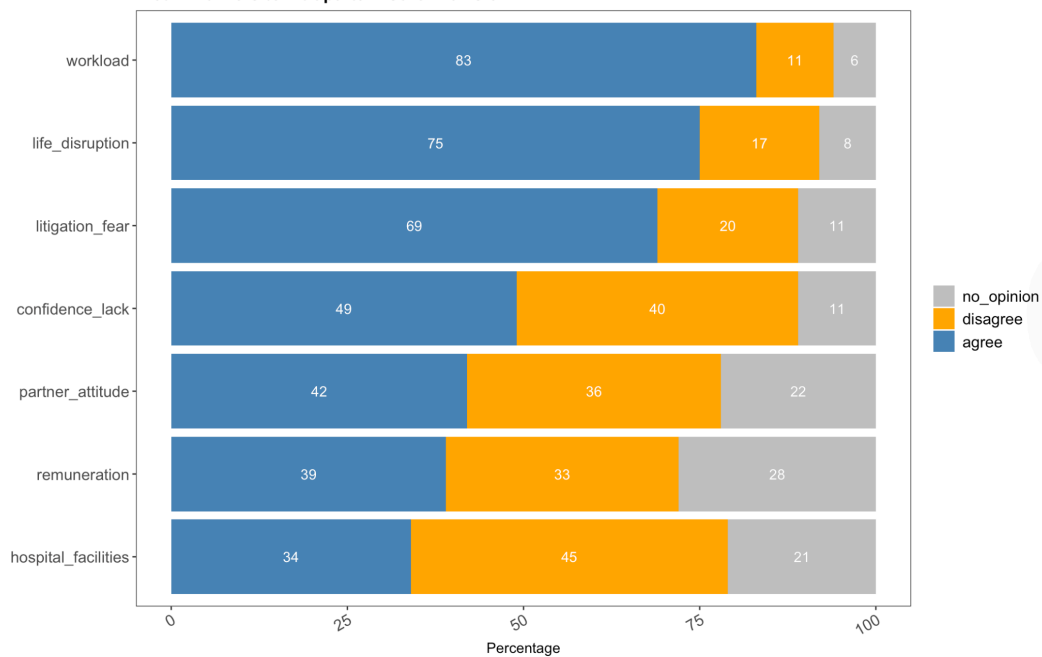
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Appendix: Graphs Revision

Plot 1: PGs Competency and Readiness in Intrapartum Care Provision



Plot 2: Barriers to Intrapartum Care Provision



Plot 3: Policy and Regulations as a Most Pronounced Aspects Defining the Provision of Intrapartum Care

